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Fraud detection using fraud triangle risk factors

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Abstract The objective of this study is to identify the financial statement fraud factors and rank the relative importance. First, this study reviews the previous studies to identify the possible fraud indicators. Expert questionnaires are distributed next. After questionnaires are collected, Lawshe's approach is employed to eliminate these factors whose CVR (content validity ratio) values do not meet the criteria. Further, the remaining 32 factors are reviewed by experts to be the measurements suitable for the assessment of fraud detection. The Analytic Hierarchy Process (AHP) is utilized to determine the relative weights of the individual items. The result of AHP shows that the most important dimension is Pressure/Incentive and the least one is Attitude/rationalization. In addition, the top five important measurements are "Poor performance", "The need for external financing", "Financial distress", "Insufficient board oversight", and "Competition or market saturation". The result provides a significant advantage to auditors and managers in enhancing the efficiency of fraud detection and critical evaluation.

Keywords Fraud detection · Fraud triangle · Analytic hierarchy process · AHP · Data mining · Lawshe's approach

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1 Introduction

The wave of financial scandals in the era of the 21st century actually elevated the awareness of the fraud (Kerr and Murthy 2013). Financial statement fraud has cast an increasingly adverse impact on the individual investors and the stability of global economies (Zhou and Kapoor 2011). The collapse of Enron has caused about a $70 billion lost in the capital market. The Computer Security Institute reported a significant increase in financial fraud cases recently (Reddy et al. 2012). The rise of fraudulent occurrence is a serious concern for the potential investors because fraudulent financial reports create a substantial negative impact on the company's reputation as well as the market value (Hogan et al. 2008).

Financial statements are basic documents to reflect company's financial status (Beaver 1966). Fraudulent financial reports are perpetrated to increase stock prices or to get loans from banks (Ravisankar et al. 2011). Financial statement fraud detection is vital because of the devastating consequences of financial statement frauds (Ngai et al. 2011). Fraud behaviors are often subtle in the beginning (Chivers et al. 2013). Therefore, it is difficult to detect them. Regulations play an important role to emphasize the responsibility of auditors to assess the risk of fraudulent financial reporting adequately (Srivastava et al. 2009). However, detecting frauds remains difficult because of the lack of a commonly accepted definition of reasonable assurance, limitations of audit methods and the cost constraints (Spathis 2002; Hogan et al. 2008). The purpose of this study is to use the expert questionnaire to filter out the critical fraud factors, and then use Analytic Hierarchy Process (AHP) to calculate weights of each individual measurement item and rank the importance of these fraud factors. This study investigates all aspects of the fraud triangle and uses the public information to measure the attributes, pressure/incentive, opportunity, and attitude/rationalization.

This research complements the previous studies in the following three ways. First, most previous technology-based tools studies solely rely on employing the past data of fraudulent companies and the related financial ratios to construct the model. The number of the cases of fraudulent companies are rather limited which may reduce the inherited advantage of such technology-based tools as data mining. Further, most previous financial statement fraud detection studies applying the data mining techniques utilize less than 100 fraud case companies to analyze fraud factors (Green and Choi 1997; Kirkos et al. 2007). In other words, these technology-based tools in auditing can't be used to screen all aspects of fraud factors and thus may provide various incompatible results given the different nature and type of tools. Secondly, the financial statement fraud factors identified in previous studies are unavailable to other users because of the data privacy. To resolve this challenge, the authors use the available and public information to define the fraud factors which can certainly provide a contribution to conduct future studies in this subject area. Thirdly, this study uses AHP to calculate the weight of each individual factor which can be deems as another unique contribution since only few studies ever investigate the rank of financial statement fraud factors. In fact, previous studies may place focus on just identifying the fraud factors. Nevertheless, to rank the significance of the fraud risk factors becomes such a critical issue since the limited budgets are always one of the main concerns encountered by today's businesses. From the above discussion, our paper applies AHP to rank the importance of financial statement fraud factors to provide the solutions to meet these aforementioned challenges including the limited budget and restricted resources. The rank of the importance of financial statement frauds may thus provide a significant advantage to both auditors and managers in enhancing the efficiency of fraud detection and performing critical evaluation.

The remainder part of the paper is organized as follows. Section 2 describes the fraud triangle and financial fraud detection and reviews the literature. In section 3, this paper presents the research flow and discusses the research methodology. This paper further analyzes the experimental results in section 4 and the last section concludes the results and provides suggestions and implications.

2 Literature review

2.1 Financial statement fraud

Financial fraud is a critical problem for external auditors because of the potential legal liability arising from the failure to detect false financial statements and the possible damage to professional reputation resulting from the public's dissatisfaction. Over the past decades, several major corporate scandals incurred in the United States, including Enron Corp., Tyco, and WorldCom Inc. The 2005 biennial survey of more than 3000 corporate officers in 34 countries, conducted by PricewaterhouseCoopers (PwC), reveals that in the post-Sarbanes-Oxley era, more financial fraud cases have been discovered and reported, as evidenced by a 140 % increase in the discovered number of financial misreporting (PricewaterhouseCoopers 2007).

2.2 Financial statement fraud factors

The fraud triangle is a model for explaining the factors that cause someone to commit the occupational fraud. It is composed of three components, which together lead to the possible fraudulent behaviors. In other words, a common model that brings these aspects together is the Fraud Triangle (Cressey 1973). In specific, this aforementioned model is built on the premise that fraud is likely to result from a combination of three factors which are motivation, opportunity and rationalization. First, there has to be an "incentive" or a "pressure" to commit fraud. To commit fraud, employee might be in financial or other types of needs. Secondly, the circumstances may be conducive to provide an "opportunity" for fraud to be perpetrated. Fraud is more likely in these companies with the following situations such as the weak internal control system, poor security over company property, or unclear policies with regard to acceptable behavior. Finally, there may be an "attitude" or "rationalization" for committing a fraud. Many people obey the law simply because they believe in it and/or they are afraid of being shamed or rejected by people they care about if they are caught.

Statement on Auditing Standards No. 99 notes that three aspects of the fraud triangle can be generally observed when the fraud occurs. There is a significant amount of studies focusing on the characteristics of fraud firms, providing the analytical supports for the fraud triangle classifications and creating a list of "red flags" used in both SAS No. 82 and SAS No. 99 (Hogan et al. 2008). These results have been widely used by professionals as a useful theoretical model and reference to explain why most frauds occur (Farahmand, and

Spafford 2013). However, only very few of these prior studies investigate all aspects of the fraud triangle. Hernandez and Groot (2007) find that incentives and opportunities for generating a fraudulent behavior are in fact associated with the higher fraud risk assessments by various audit partners, and the most important factors are senior management ethical attitudes and dishonest. Rezaee (2005) justifies the existence of three fraud triangle conditions in the fraud firms. Lou and Wang (2009) examine the fraud factors and the core of fraud auditing standards to assess the possible likelihoods of the fraudulent financial statements.

However, many other studies focus on just one of the three aspects of the fraud triangle. For example, Dechow et al. (1996) find that an important motivation to manipulate the earnings is the desire to attract the external financing at a low cost. Gillett and Uddin (2005) uncover that the attitude of the CFO toward the behavior of fraudulent reporting tends to be a major influence on the intention of misreporting. Suyanto (2009) constructs the model to predict the likelihood of fraudulent financial reports based on the fraud triangle but only discusses the factors of pressures and opportunities associated with the financial statement frauds.

In the next section, this study reviews the academic findings related to the presence of these three aforementioned conditions occurred in the financial statement fraud.

2.2.1 Pressure/incentive

Incentive or motivation to commit fraud results from a perceived pressure on a person. Pressure constitutes an incentive to act in a given manner. The incentives to misstate earnings are from the pressure to meet analysts' forecasts, the compensation and incentive structures, the need for external financing, and/or the poor performance of the organization (Hogan et al. 2008). In terms of opportunity, fraud is more likely in companies with weak internal control systems, poor security over company property, or unclear policies with regard to acceptable behavior.

The desire to meet analysts' forecasts is one of the external incentives (Kaplan 2001). The differences between financial analysts' forecasted earnings and reported earnings are positively related with the likelihood of frauds. Young (2000) finds that to meet the external analyst's forecasts is the motivation to commit frauds in 43 % of cases. Lou and Wang (2009) show that the fraudulent report is positively correlated to analysts' forecast errors. This study uses analysts' forecast errors and defines it further as analysts' forecasts vs. actual measure to assess one of the financial pressures on/to management.

Stock options are associated with stronger incentives to misreport because options may link to CEO wealth with stock price (Burns and Kedia 2006). A Glass and Co. (GLC) (2005) report states that about half of the companies implicated in backdating their stock options have also restated their financial statements. Goldman and Slezak (2006) show that performance based compensation can induce managers to misreport performance. Therefore, the higher the correlation between the CEO compensation and reported income, the more income-increasing accounting choices a CEO may make, hence the higher the probability of fraud will occur. This study use CEO Pay-Performance Sensitivity to measure the incentive to commit the fraud.

In Taiwan, the Taiwan Economic Journal (TEJ) creates and releases the Taiwan Corporate Credit Risk Index (TCRI) for all listed companies. TCRI provides a comprehensive evaluation of each firm's profitability, security, operations, size, and business and environmental risks on a 10-point scale, with a larger point indicating a higher risk. This basic ranking is further refined by considering each company's pressure index that is determined by a firm's capability to manage its assets, firm's incapability of paying loan interest for more than 3 years, and the difficulty of short-term credit expansion (Yu et al. 2009). We use debt/equity ratio and TCRI index to measure the pressure on firm's debt covenant. We also follow the study of Dechow et al. (1996) to include the factor, the need for external financing, which is "capital expenditures/current assets.

Firms with a superior or satisfactory performance are motivated to falsify the financial reports, whereas poorly performing firms are motivated to falsify the reports. Bell et al. (1991) suggest that the poor financial performance increases the likelihood of financial frauds. We use general indicators of performance including return on assets, return on equity, Tobin's q and operation revenue growth rate. If a firm ranks lower than that of industry median, it has a poor performance pressure. Another poor performance measure is adopted is to use at least two annual net losses and two annual negative cash flows from the firm's operations in the research period (DeAngelo and DeAngel 1990; Beasley 1996).

Financial distress may be an incentive factor for management to commit fraud (Zhou and Kapoor 2011; Huang et al. 2012; Kirkos et al. 2007). Prior research shows that fraud-discovered companies have a greater average level of financial distress than non-fraud companies in the same control sample (Bell et al. 1991; Ravisankar et al. 2011). The measurement of a company's financial condition is operationalized by using ZFE score predict model (Carcello and Neal 2000). Another measurement of financial distress is whether the firms received an ongoing concern before the event year.

Other incentives or pressures may result when financial stability or profitability is threatened by economic, industry, or entity's operating conditions. Examples of incentives include such items as a high level of competition or the market saturation with declining profit margins. Persons (1995) argues that management of fraudulent firm is more competitive than that of non-fraudulent firms in using the firm's assets to generate sales. From the above discussion, we choose Capital turnover (Sales/Total Assets) as a proxy for the competitive situations of a firm (Persons 1995).

2.2.2 Opportunity

Opportunity is the condition or situation that would allow a person to commit fraud. For example, the lack of sufficient board oversight may allow the management to inappropriately manipulate the reported earnings to reach the analysts' forecasts (Wilks and Zimbelman 2004). SAS No. 99 provides examples of risk factors that increase the opportunity to commit fraudulent financial reporting. These risk factors include the nature of the industry or the entity's operations (e.g. significant, complex or related party transactions), ineffective monitoring of management, a complex organizational structure (e.g. one that involves several legal entities), and ineffective controls due to a lack of monitoring of controls or circumvention of controls.

Complex organizations and transactions lead to the increased opportunity for subjective interpretation, borderline disclosure and even misrepresentation (Huang et al. 2008). For an entity's operations on significant complex or related party transactions, we use the percentage of sales-related party transaction, the percentage of purchases-related party transaction, the percentage of loans-related party transaction, and the ratio of the assurance of related-party to measure the opportunity factors (Chen and Elder 2007).

The composition of the board will affect the effectiveness of the board (Fama and Jensen 1983). Factors measuring the supervision effectiveness on boards include the percentage of independent members on the board of directors, the percentage of outside members on the board of director, the percentage of institutional ownership, percentage of management ownership, percentage of blockholders ownership, abnormal change of board members, consecutive changes in insider holdings, board size, family firm, and the deviation between voting and cash flow rights (Hadani 2012; Suyanto 2009; Fama and Jensen 1983; Beasley 1996; Millar and Yeager 2007; Anderson and Reeb 2003).

Weak internal control environments may provide opportunities to engage in fraudulent financial reporting. Internal control over financial reporting has long been recognized as an important feature of a company (Kinney 2005). We use significant deficiencies of the internal control, turnover frequency of internal auditor (Hammersley et al. 2012; Lou and Wang 2009), and firm size to measure the monitoring effectiveness. In addition, we use the inventory-to-total-assets ratio, and the auditor tenure to firm age to measure the opportunity of fraud.

2.2.3 Attitude/rationalization

Those involved in fraudulent financial reporting are usually capable of rationalizing fraudulent acts as being consistent with their personal code of ethics (Suyanto 2009). Zandstra (2002) indicates that the central reason for Enron's demise was a failure of the board of directors to function in a morally and ethically responsible manner. Apostolou et al. (2001) examine that management characteristics and influence over the control environment are approximately twice as important as operating factors, and financial stability red flags are about four times as important as industry conditions. Therefore, the management characteristics and ethics are a major determinant to attitude or rationalization.

The commission of fraud at the highest levels in an organization ultimately involves with an individual's decision to participate in or to acquiesce (Zahra et al. 2007). Zahra et al. (2007) investigate the effect of management characteristics on/to fraud. Some other studies have shown that adequate business and economics education may actually cause a decline in moral behavior, because such programs can increase self-interested behaviors in some individuals and thereby encourage unethical practices and fraud. For this reason, we use education of top management to measure attitude/rationalization. Moreover, we also use top manager's education background to measure attitude/rationalization.

Management fraud has been found to be typically committed by top management. From a legal perspective, firms with executives who have ignored, condoned, rewarded, and/or participated in past instances of wrongdoing, will likely be recidivists due to the predisposition of their behavior and attitude. From a legal point of view, managers who are dishonest and are evasive towards their auditors, are more likely to engage in financial fraud (Loebbecke et al. 1989). The ethics and conduct of senior managers in a corporation set the overall, ethical tone in an organization (Hernandez and Groot 2007). Therefore, we measure the firm's ethical culture by using number of indictment on civil or criminal litigation.

Audits are valued by investors because they assure the reliability of, and reduce the uncertainty associated with, financial statements (Francis and Wilson 1988). Hiring an auditor with a high-quality reputation can be a means to reduce agency costs (Francis and Wilson 1988). In addition, hiring an auditor with a high-quality reputation also helps management to reassure investors that a company's financial statements are reliable and accurate (Kinney 2005). Further, auditors sanctioned by regularity authorities have performed the lower audit quality. Consequently, we use whether the auditor was sanctioned by regularity authorities and the frequency of restatement as the proxy of rationalization or attitude.

Management manipulates earnings manipulation for hiding deteriorating performance (Stratopoulos et al. 2013). Several studies suggest that an undue emphasis on earnings projections and aggressive management attitudes toward financial reporting are primary indicators of fraudulent financial reporting (Bell and Carcello 2000; Loebbecke et al. 1989). Earnings management is the most common method of engaging in fraudulent financial reporting. Fraudulent financial reporting can be either the direct falsification of transactions or intentional delay (early) recognition of transactions that eventually occur (Rezaee 2005). From the above discussion, we use performance-matched discretionary accruals, introduced in the study of Kothari et al. (2005), to measure earnings manipulation.

Chen and Elder (2007) explore some significant fraud risk factors, such as the number of earnings restatements, the number of external auditor switches, and the number of internal auditor switches, as proxies for Rationalization or Attitude. Gillett and Uddin (2005) conclude that the attitude of the CFO toward the behavior of fraudulent reporting to be a major influence on/to intention to misreport. Agrawal et al. (2006) prove that firms suspected of or charged with fraud have unusually high turnover among senior managers and/or directors. Therefore, we use CEO turnover frequency and CFO turnover frequency to proxy for rationalization or attitude.

TSAS No.43 indicates that auditors may become aware of such information that may indicate a risk factor and that is the relationship between management and the current or predecessor auditor is strained, such as frequent disputes with the current or predecessor auditor about accounting, auditing, or reporting matters. Knechel et al. (2007)) suggest that clients may use auditor switches to lower the likelihood of detection of fraudulent financial reporting. For this reason, we use CPA turnover frequency to proxy for rationalization or attitude.

Loebbecke et al. (1989) find that 75 % of observations are the indication of decision-making domination by one person or small group acting in concert and these are weak internal controls. TSAS No.43 suggests non-separating ownership and control is one of fraud factors for rationalization or attitude. Consequently, we use CEO duality and a director occupying the CEO position to proxy for rationalization or attitude.

2.3 Financial statement fraud detection

Financial statement fraud detections have become an increasingly demanding task. However, detecting financial fraud using traditional audit procedures is a rather difficult task and there may be several reasons. First, there is a shortage of needed knowledge concerning the characteristics to manage the relevant fraud. Secondly, as the fraudulent manipulation of accounting data is so infrequent, most auditors may lack the sufficient experience and background needed to detect it in an effective manner. Finally, managers may deliberately try to deceive the auditors (Fanning and Cogger 1994). While knowing the limitations of an audit, finance and accounting managers have concluded that the traditional and standard auditing procedures are insufficient to detect frauds.

A fruitful area of prior research has been placed on the tools and techniques such as analytical procedures, ratio analysis, regression analysis, and checklists to improve the fraud detection (Hogan et al. 2008). Bell and Carcello (2000) investigate whether a logistic regression model including some significant risk factors can perform well in predicting the fraud and this study uncover that a simple logistic model may outperform the auditors in terms of the fraud risk assessment. Spathis (2002) construct a model to detect the falsified financial statements. In fact, this study employed the statistical method of logistic regression and the obtained results suggest that there is a potential in detecting the falsified financial statements through the analysis of published financial statements.

Data Mining is regarded as an iterative process with which the progress can be defined by the discovery of relationships, either through automatic or manual methods. Based on the data mining techniques, a response model needs to be built as a decision model for handling the prediction or the classification of a potential domain problem like expert systems (Lu and Chen 2009). Loebbecke et al. (1989) conduct an experiment to examine if the use of an expert system can be employed to enhance the auditors' performance. In addition, neural networks may constitute one of the most widely used techniques in data mining area (Chen et al. 2009). In fact, Green and Choi (1997) develop a Neural Network fraud classification model and this model used five ratios and three accounts as an input. Their results show that Neural Networks have the significant capabilities to deal effectively with the fraud cases when they are used as a fraud detection tool. Finally, Chen et al. (2009) show empirically that a neural network may provide not only a promising prediction accuracy and a better detecting power but also incur a lower misclassification cost.

Another approach to identify the financial fraud is the application of Benford's Law. Benford's Law has been promoted to provide the auditors as a tool/technique that is simple and effective for detecting the fraud (Durtschi et al. 2004). The purpose of this technique is to be utilized to assist the auditors in reviewing the overwhelming volumes of datasets and then identifying any potential fraud records. Cleary and Thibodeau (2005) examine whether the digital analysis using Benford's Law technique has any merit as a fraud detection tool, and this study finds that using the digit-by-digit approach may actually increases the chance of a Type I error, but it also tends to increase the chance of finding a fraud. Huang et al. (2008) propose an innovative fraud detection mechanism based on Zipf's law to develop an innovative fraud detection. The simulation experimental results in this aforementioned study demonstrate that Zipf

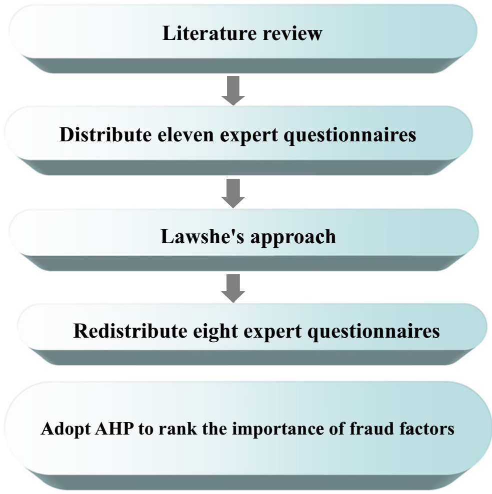

Fig. 1 Research flow

Analysis can be employed to assist auditors to locate the source of suspicion and enhance further the efficiency of the resulting audit processes.

Nonetheless, most of these aforementioned technology-based tools studies rely on the past data of fraudulent companies and the financial ratios to construct fraud risk model. The cases of fraudulent companies are limited which reduce the advantage of these technology-based tools. Previous financial statement fraud detection studies applying technology-based techniques utilize less than 100 fraud case companies to analyze (Green and Choi 1997; Kirkos et al. 2007). Also, these technology-based tools in auditing can't be used to screen all aspects of fraud factors and may provide various incompatible results given the different nature and type of tools. Because of these limitations, it is the authors' attempt to use experts' experience to identify frauds and rank these frauds to assist auditors and investors to prevent and detect the frauds. This study adopts AHP which uses a specific input format for decision makers to express their preferences regarding multiple criteria and alternatives namely, pairwise comparisons (Benítez et al. 2012).

3 Research design

3.1 Research flow

The research flow presents in Fig. 1. First, fifty-three fraud factors are identified and selected by reviewing the previous studies in the literature review section. Then, after eleven expert questionnaires are distributed and collected, we employed Lawshe's approach to filter out thirty-two critical fraud factors. This aforementioned process extracts and filters the fraud factors to create a list that is suitable for a further exploration and examination. Finally, we redistributed eight expert questionnaires with thirty-two fraud factors in twenty categories and adopted AHP technique to compute the weights of individual measurement to rank the measurement in each of three dimensions of the fraud triangle.

3.2 Expert questionnaire - Lawshe's approach

This study summarizes fraud factors and categorizes them into three dimensions of fraud triangle. To ensure and enhance the content validity of individual dimension and associated items, as well as to remedy the insufficient findings from prior literature and meet the practical needs, this paper follows the research methodology and validation process proposed by Lawshe (1975). This paper conducts the experts' questionnaire survey to confirm the suitability of the fraud factors. This process extracts and filters the fraud factors suitable for the exploration of the fraud detection. In addition, all the experts are invited from both the industries and academic communities.

The expert questionnaires have three parts. The first part is the questionnaire explanation, which elucidates the content and purpose of this research. The second part is a number of questions, which first request the experts to provide their basic information and check the significance of critical influential factors in each aspect. Here, this study uses three-point scale that is 1 as not relevant, 2 as important (but not essential), and 3 as essential. The Lawshe's approach is applied to justify the

Table 1 Basic information of experts for Lawshe's analysis

Group	Type	Number	Positions	Average years of service
Academia	Professor	3	Department of accounting*2	12
			Department of accounting and information technology*1	9
Industry	Companies	4	Audit, Manager*2	11
			Accounting, Manager*2	14
	CPA firms	4	Auditing, Manager *3	6
			IT Auditing, Manager*1	4

Table 2 Result of expert questionnaires - Lawshe's approach

Dimension	Category	Factor	CVR value	CVR value > 059
Pressure/ Incentive	**Meet analysts' forecasts**	**Meet analysts' forecasts**	**0.64**	**Yes**
	The need for external financing	**The need for external financing _Taiwan Corporate Credit Risk Index**	**1.00**	**Yes**
		Compensation and incentive structures _CEO Pay-Performance Sensitivity	0.45	No
		The need for external financing _debt/equity ratio	**1.00**	**Yes**
		The need for external financing _(cash from operations – mean capital expenditures)/current assets	**0.82**	**Yes**
	Poor performance	**Poor performance _ return on assets**	**1.00**	**Yes**
		Poor performance _return on equity	**0.64**	**Yes**
		Poor performance _Tobin's q	0.45	No
		Poor performance _at least two annual net losses	**0.64**	**Yes**
		Poor performance _at least two annual negative cash flows from operations	**0.64**	**Yes**
	Financial distress	**Financial distress _going concern opinion**	**1.00**	**Yes**
		Financial distress _the higher probability of bankruptcy	**0.82**	**Yes**
	Competition or market saturation	**Competition or market saturation _higher growth**	**0.64**	**Yes**
		Competition or market saturation _Capital turnover	0.45	No
Opportunity	**Related party transaction**	**Related party transaction _percentage of sales-related party transaction**	**1.00**	**Yes**
		Related party transaction _percentage of purchases-related party transaction	**0.82**	**Yes**
		Related party transaction _the ratio of the assurance of related- party	**0.64**	**Yes**
		Related party transaction _percentage of loans-related party transaction	0.45	No
		Related party transaction _the ratio of the assurance of related- party	0.45	No
	Complex transactions	**Complex transactions _equity investment ratio**	**0.64**	**Yes**
		Complex transactions _foreign re-investment	**0.64**	**Yes**
		Complex transactions _the percentage of sales which are foreign	**0.64**	**Yes**
		Complex transactions _issue ECB	0.27	No
	Insufficient board oversight	**Insufficient board oversight _family firm**	**0.64**	**Yes**
		Insufficient board oversight _consecutive changes in insider holdings	**1.00**	**Yes**
		Insufficient board oversight _deviation between voting and cash flow rights	**0.64**	**Yes**
		Insufficient board oversight _Abnormal change of board members	0.45	No
		Insufficient board oversight _pyramid structures	0.27	No
		Insufficient board oversight _cross-holdings	0.27	No
		Insufficient board oversight _percentage of independent members on the board of director	−0.09	No
		Insufficient board oversight _percentage of outside members on the board of director	−0.82	No
		Insufficient board oversight _percentage of institutional ownership	−0.82	No
		Insufficient board oversight _percentage of management ownership	**0.64**	**Yes**
		Insufficient board oversight _percentage of blockholders ownership	**0.64**	**Yes**
		Insufficient board oversight _board size	−0.82	No
	Internal control environment	**Internal control environment _internal control statement about significant deficiencies**	**0.82**	**Yes**
		Internal control environment _turnover frequency of internal auditor	**0.82**	**Yes**
		Internal control environment _firm age	**0.64**	**Yes**
		Internal control environment _firm size	0.27	No
		Internal control environment _auditor tenure	−0.09	No
	Other opportunity	**Other opportunity _audited by a Big 4**	**0.64**	**Yes**
		Other opportunity _inventory/total assets ratio	0.45	No
Attitude/ rationalization		Management characteristics _education of top management	0.27	No
		Management characteristics _top manager's education background	−0.09	No

Table 2 (continued)

Dimension	Category	Factor	CVR value	CVR value > 059
	Ethic	Ethic _ number of indictment on civil or criminal litigation	0.45	No
		Ethic _Historical restate frequency	**0.82**	**Yes**
		Engagement CPA who were sanctioned by regularity authorities	0.45	No
	Earning manipulation	**Earning manipulation**	**0.82**	**Yes**
	Top management turnover	**Top management turnover _CEO turnover frequency**	**0.82**	**Yes**
		Top management turnover _CFO turnover frequency	**0.82**	**Yes**
	CPA turnover frequency	**CPA turnover frequency**	**1.00**	**Yes**
	Non-separating ownership and control	**Non-separating ownership and control _CEO duality**	**0.64**	**Yes**
		Non-separating ownership and control _a director occupying the CEO position	0.45	No

content validity of each factor. This assessment approach of content validity is basically utilized to ensure the representativeness and adequacy of the factors in the specific domain (Lewis et al. 1995).

According to Lawshe, this study eliminates the items for which content validity ratio (CVR) are less than a number which is based on the formula to calculate the content validity ratio CVR for each item:

$$\mathbf{CVR} = (\mathbf{n}-\mathbf{N}/\mathbf{2})/(\mathbf{N}/\mathbf{2}) \tag{1}$$

Where n is the frequency count of the number of committee members rating the item either "3—essential" or "2—important (but not essential)", and N is the total number of panelists.

3.3 Experts' questionnaire - AHP

To determine the relative weights of each individual item, this paper designs a hierarchical structure of questions using AHP based on the measurement dimensions and items derived from above. To make a good decision requires judgments of what is more likely or more preferred over different time periods. There are at least three ways to deal with the dynamic decisions-making. One alternative is to include different factors in the structure itself to indicate the possible changes in time such as scenarios and different time periods and then carry out the paired comparisons with respect to these aforementioned periods using the fundamental scale of the AHP (Saaty 1980). Based on pairwise comparison judgments, AHP integrates both important criteria and alternative preference measures into a single overall score for ranking different decision alternatives (Ngai 2003).

AHP uses a specific input format for decision makers to express their preferences regarding multiple criteria and alternatives, namely, pairwise comparisons (Benítez et al. 2012). The scale is divided into nine levels, which are equally important, slightly important, more important, very important, and significantly important. They are given the weights of 1, 3, 5, 7, and 9 respectively. Also, among the five basic measurement rules, there are four measurements with a value of 2, 4, 6, and 8 accordingly. Markers close to the right mean that the right-side factors are more important than the factors on the left side. The respondents are requested to check the most appropriate

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Group	Type	Number	Positions	Average years of service
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			Department of accounting and information technology*1	9
Industry	Companies	3	Audit, Manager*2	8
			Accounting, Manager*1	13
	CPA firms	3	Auditing, Manager *2	10
			IT Auditing, Manager*1	6

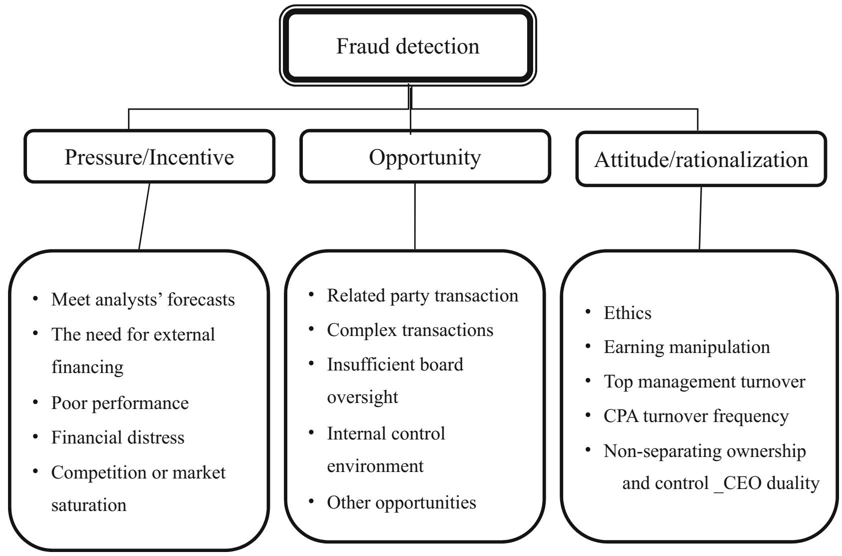

Fig. 2 Decision hierarchies

level based on their experience and opinion to each question. After obtaining the questionnaire data, AHP is used to compare the importance of factors by pairing in each subsystem.

4 Empirical results

4.1 Results of experts' questionnaire

4.1.1 Lawshe's approach

The expert questionnaires are distributed to the academia, industry practitioners, and CPA firms through e-mail so that the experts can provide a direct response accordingly. There are eleven experts and the demographic information of the selected experts is provided in Table 1 below.

In Table 2, since there are eleven experts in this study, the Content Validity Ratio (CVR) should be greater than 0.59 to be selected (Lawshe 1975). This paper eliminates 21 factors whose CVR values do not meet the criteria. As a result, the remaining 32 factors are suitable measurements for the assessment of fraud detection. The results of the expert questionnaires are provided in Table 2. The fraud factors that meet the criteria (CVR > 0.59) are shown in bold in Table 2 and selected for a further AHP analysis. In summary, there are 11 out of 32 factors belonging to pressure/incentive dimension, 15 out of 32 factors belonging to the opportunity dimension, and finally 6 out of 32 factors belonging to the attitude/rationalization dimension. In the pressure/incentive dimension, "Meet analysts' forecasts", "The need for external financing _Taiwan Corporate Credit Risk Index", "The need for external financing _debt/equity ratio", "The need for external financing _(cash from operations – mean capital expenditures)/current assets", "Poor performance _ return on assets", "Poor performance _return on equity", "Poor performance _at least two annual net losses", "Poor performance _at least two annual negative cash flows from operations", "Financial distress _going concern opinion", "Financial distress _the higher probability of bankruptcy", and "Competition or market saturation _higher growth" are selected. As for the opportunity dimension, "Related party transaction _percentage of sales-related party transaction", "Insufficient board oversight _consecutive changes in insider holdings", "Related party transaction _percentage of purchases-related party transaction", "Related party transaction _the ratio of the assurance of related- party", "Complex transactions _equity investment ratio", "Complex transactions _foreign re-investment", "Complex transactions _the percentage of sales which are foreign", "Insufficient board oversight _family firm", "Insufficient board oversight _deviation between voting and cash flow rights", "Insufficient board oversight _percentage of management ownership", "Insufficient board oversight _percentage of blockholders ownership", "Internal control environment _internal control statement about significant deficiencies", "Internal control environment _turnover frequency of internal auditor", "Internal control environment _firm age", and "Other opportunity _audited by a Big 4" are identified. Finally, in the Attitude/rationalization dimension, "Ethic _Historical restate frequency", "Earning manipulation", "Top management turnover _CEO turnover

Table 4 Relative weights of three dimensions

Measurement Dimensions	Hierarchical weight %	Weight ranking
Goal: Fraud Detection		
Pressure/Incentive	55.7	1
Opportunity	28.5	2
Attitude/rationalization	15.8	3

Table 5 Relative weights of pressure/incentive

Dimensions	Measurements	Level weight %	Overall weight %	Ranking
Goal: Fraud Detection - Pressure/Incentive				
Pressure/Incentive	Poor performance	32.5	18.1	1
	The need for external financing	25.2	14.0	2
	Financial distress	20.4	11.3	3
	Competition or market saturation	12.3	6.8	4
	Meet analysts'forecasts	9.6	5.3	5

frequency", "Top management turnover _CFO turnover frequency", "CPA turnover frequency", and "Non-separating ownership and control _CEO duality" are the definite choices.

After conducting the expert questionnaires' calculation and matching of CVR values, it was notable that in the dimension of pressure/incentive, only one category - "Compensation and incentive structures" is eliminated. In the dimension of attitude/rationalization, all categories are kept except the following two - "Management characteristics (two factors)", and "Engagement CPA who were sanctioned by regularity authorities".

4.1.2 Analytic hierarchy process (AHP)

From the results of Lawshe's Approach, the remaining 32 factors are considered the measurements suitable for the assessment of fraud detection. In summary, 11 out of 32 factors are included in five categories such as "Meet analysts' forecasts", "The need for external financing", "Poor performance", "Financial distress", and "Competition or market saturation" and these five categories fall in pressure/incentive dimension. Further, 15 out of 32 factors are in five categories including "Related party transaction", "Complex transactions", "Insufficient board oversight", "Internal control environment", and "Other opportunities". These five categories are in the opportunity dimension. Finally, 6 out of 32 factors are in five categories "Historical restate frequency", "Earning manipulation", "Top management turnover", "CPA turnover frequency", and "Non-separating ownership and control" and these five categories are in the attitude/rationalization dimensions. This study categorizes 32 factors and redistributes eight expert questionnaires for further AHP analyzation. The demographic information of eight experts is provided in Table 3 below. For designing the paired comparison matrices, the decision hierarchies are formed as depicted in Fig. 2. In each dimension, five categories are selected from the Lawshe's Approach from the aforementioned step. Moreover, each level in the hierarchy corresponds to the common characteristic of the elements in that level. This paper formulates an appropriate AHP model consisting of the Goal, Issues, and Categories.

To ensure the consistency of paired questions answered by the eight experts, this paper run the consistency tests. This paper sets Consistency Ratio (C.R.) <0.1 as the maximum acceptable error (Saaty 1980). If the C.R. < 0.1, it is deemed as a valid questionnaire. On the other hand, if the C.R. >= 0.1, it is deemed as an invalid questionnaire. In second questionnaires distribution, the total of eight experts passed the consistency tests. Therefore, this paper performs the AHP analysis on these eight valid questionnaires to calculate further the overall and local weighs of their corresponding factors.

Relative weights of three dimensions As shown in Table 4, among the three dimensions for fraud triangle, the dimensions for "Pressure/Incentive" accounts for the highest weight (55.7 %), with "Opportunity" (28.5 %) next, and the lowest weight (15.8 %) goes to "Attitude/rationalization".

Relative weights of each category from three dimensions After deriving the relative weights of the three dimensions,

Table 6 Relative weights of opportunity

Dimensions	Measurements	Level weight %	Overall weight %	Ranking
Goal: Fraud Detection - Opportunity				
Opportunity	Insufficient board oversight	30.2	9.3	1
	Related party transaction	21.9	6.8	2
	Complex transactions	20.0	6.7	3
	Internal control environment	16.4	6.2	4
	Other opportunity	11.4	3.5	5

Table 7 Relative weights of attitude/rationalization

Dimensions	Measurements	Level weight %	Overall weight %	Ranking
Goal: Fraud Detection				
- Attitude/rationalization				
Attitude/rationalization	CPA turnover frequency	37.8	5.1	1
	Historical restate frequency	17.9	2.4	2
	Earning manipulation	13.2	1.8	3
	Top management turnover	9.7	1.3	4
	CEO duality	7.0	1.0	5

this paper further analyzes and explains the weights of the respective measurements of each dimension. Level weights can also be referred as local priority, which is the relative compared weight between the standards of each level. The other overall weight is also called the global priority. This multiplies the upper level weight by the relative weights for each principle, which is used to show the weight of the principles in this level of the overall evaluative mechanism.

Table 5 indicates the weights of "Pressure/Incentive" dimension. The category "Poor performance" is the most important one in the fraud detection, which accounts for 32.5 %. Second important category in fraud detection is "The need for external financing".

Table 6 indicates the weights of "Opportunity" dimension. The category "Insufficient board oversight" is the most important one in the fraud detection, which accounts for 30.2 %. This result is consistent with prior research that the board of directors can be deemed as the highest internal control mechanism since it is responsible for monitoring the actions of top management (Fama and Jensen 1983).

Relative weights of all categories From the overall point, Table 8 indicates the weights of all categories. The top five important categories are "Poor performance (18.1 %)", "The need for external financing (14 %)", "Financial distress (11.3 %)", "Insufficient board oversight (9.3 %)", "Competition or market saturation (6.8 %)". These five categories, except "Insufficient board oversight", are all from Pressure/Incentive dimension. In addition, the lowest weight is in Attitude/rationalization dimension.

The Fraud Triangle consists of three dimensions (Incentive/ Pressure, Opportunity, and Attitude/ Rationalizations) when fraud occurs. The results of AHP can find the weight from three different dimensions of fraud triangle. The most important dimension is Pressure/Incentive, next dimension is Opportunity, and the lowest dimension is Attitude/ rationalization.

Table 7 indicates the weights of "Attitude/rationalization" dimension. The category "CPA change" is the most important factor in the fraud detection, which accounts for 37.8 %. This category is the most important in Opportunity dimension by

Table 8 Relative weights of all categories (Analyzed in three independent dimensions)

Dimensions	Measurements	Overall weight %	Ranking
Goal: Fraud Detection			
Pressure/Incentive	Poor performance	18.1	1
Pressure/Incentive	The need for external financing	14.0	2
Pressure/Incentive	Financial distress	11.3	3
Opportunity	Insufficient board oversight	9.3	4
Pressure/Incentive	Competition or market saturation	6.8	5
Opportunity	Related party transaction	6.7	6
Opportunity	Complex transactions	6.2	7
Pressure/Incentive	Meet analysts' forecasts	5.3	8
Opportunity	CPA turnover frequency	5.1	9
Opportunity	Internal control environment	5.0	10
Attitude/rationalization	Other opportunity	3.5	11
Attitude/rationalization	Historical restate frequency	2.4	12
Attitude/rationalization	Earning manipulation	1.8	13
Attitude/rationalization	Top management turnover	1.3	14
Attitude/rationalization	CEO duality	1.0	15

the experts. TSAS No.43 indicates that the relationship between management and the current or predecessor auditor is very important.

5 Conclusions

After the issuance of SAS No. 82, auditors have more responsibility to detect, manage and control frauds. Several academic studies investigate and explore this subject area (Spathis 2002; Owusu-Ansah et al. 2002; Gillett and Uddin 2005). Numerous studies use the red flags listed in SAS No. 99 to study fraudulent financial reporting and investigate the effectiveness of those indicators to predict fraud. Few studies, however, investigate and examine all aspects of the fraud triangle. In addition, most of prior studies of the fraud triangle or red flags have the deficiencies such as using survey or subjective measure, fraud factor data unavailable to other users, and difficult to perform the empirical analysis. This study uses public information to perform the assessment and ranking. The results are available for other users to pursue further studies.

The study is designed to evaluate and rank the financial statement fraud factors. To select important factors, this study uses Lawshe's approach to conduct experts' questionnaire survey to confirm the suitability of fraud factors. From the results of Lawshe's Approach, the remaining 32 factors are considered to be the measurements suitable for the assessment of fraud detection. This study redistributes expert questionnaires for further AHP analyzation Table 8.

The AHP results indicate that the highest weight of three fraud triangle dimension is "Pressure/Incentive", next is "Opportunity", and the lowest weight is "Attitude/rationalization". The top five most important factors are "Poor performance", "The need for external financing", "Financial distress", "Insufficient board oversight", and "Competition or market saturation" by sequence. The results show that experts focus more on "Pressure/Incentive" than "Opportunity", and do not pay much attention to "Attitude/rationalization" dimension.

From the experts' point of view, firms with the poor performance and the stronger need for external financing pressure have a higher incentive to perform fraudulent activities. This result is consistent with previous findings that the management is easily motivated to engage in manipulating financial reports when the corporation have a poor operating performance or face financial difficulties (Stice 1991; Spathis et al. 2003). From the results, we learn that companies with a poor financial structure are easily involves in fraudulent behaviors.

This paper uses Lawshe's approach to confirm the suitability of the fraud factor evaluation and adopts AHP to calculate the weighs of individual measurement items to rank the importance of factors. Different from prior studies, this paper utilizes experts' professional judgment (the experts include those from the academia, industry, and CPA firms) to determine the importance of factors. The results obtained from this study can be employed as a guideline to design and develop effective and efficient control mechanism in the certain areas to reduce and minimize the risks of fraud occurrence. The limitations of this research are as follows. First, though we try collect as many fraud factors as we can, we may still miss some fraud factors. Second, this research uses the experts' questionnaire to further rank the importance of fraud factors. The personal judgments of experts may affect the results.

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